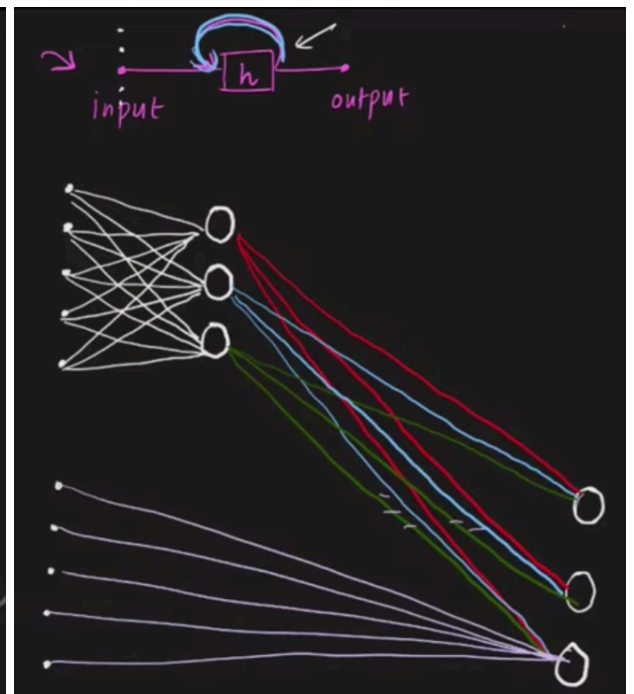
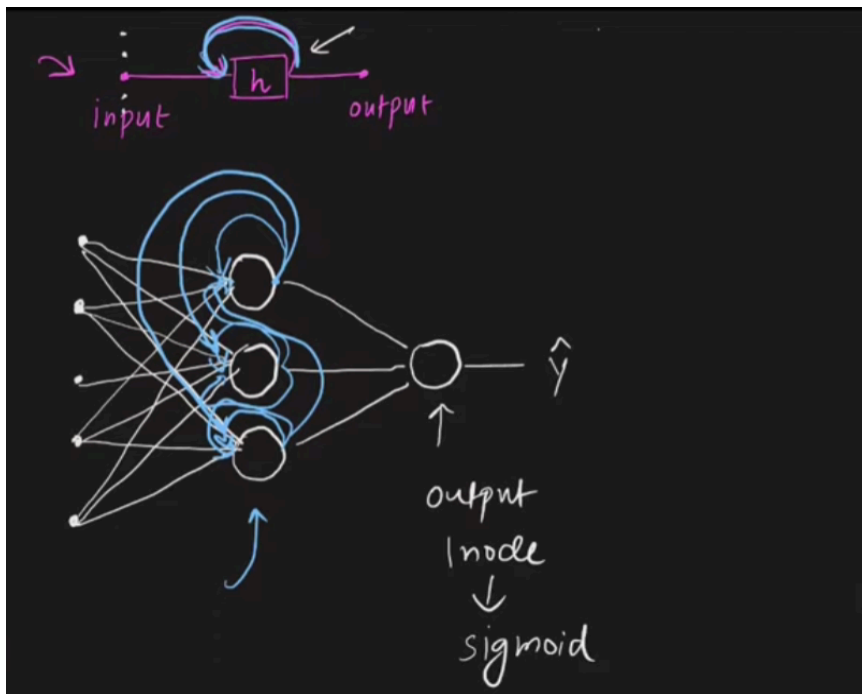


1. RNN internal structure



2. Cross-entropy Intuition

The exact mapping

For one data point, cross-entropy loss is

$$\mathcal{L} = -\log(p)$$

where

$p = \hat{y}_{\text{true}}$ = predicted probability of the **correct class**.

So the penalty is a **logarithmic function** of the confidence.

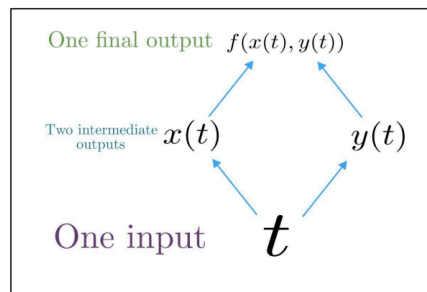
Probability → Penalty table

Predicted prob. p	Loss = $-\log(p)$	Intuition
1.0	0.000	Perfect, no penalty
0.9	0.105	Very confident, tiny penalty
0.7	0.357	Reasonable confidence
0.5	0.693	"Coin flip" uncertainty
0.1	2.303	Very wrong
0.01	4.605	Extremely wrong
$\rightarrow 0$	$\rightarrow \infty$	Impossible prediction

👉 Penalty grows fast as confidence drops.

3. Multivariable-chain rule

Multivariable Chain Rule



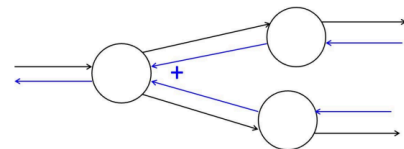
Source:
<https://www.khanacademy.org/math/multivariable-calculus/multivariable-derivatives/differentiating-vector-valued-functions/a/multivariable-chain-rule-simple-version>

- Given a multivariable function $f(x, y)$, and two single variable functions $x(t)$ and $y(t)$, here's what the multivariable chain rule says:

$$\frac{d}{dt} f(x(t), y(t)) = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

Derivative of composition function

Gradients sum at outward branches



$$a = x + y$$

$$b = \max(y, z)$$

$$f = ab$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial a} \frac{\partial a}{\partial y} + \frac{\partial f}{\partial b} \frac{\partial b}{\partial y}$$

4. Jacobian Derivation

Deriving the Jacobian

The RNN hidden state update is:

$$h^{(j)} = f(W_{hh} h^{(j-1)} + W_{hx} x^{(j)})$$

Let $z^{(j)} = W_{hh} h^{(j-1)} + W_{hx} x^{(j)}$ (the pre-activation). Then by chain rule:

$$\frac{\partial h^{(j)}}{\partial h^{(j-1)}} = \frac{\partial h^{(j)}}{\partial z^{(j)}} \cdot \frac{\partial z^{(j)}}{\partial h^{(j-1)}}$$

First term — since f is applied elementwise, the derivative of $h^{(j)}$ w.r.t. $z^{(j)}$ is diagonal:

$$\frac{\partial h^{(j)}}{\partial z^{(j)}} = \text{diag}[f'(z^{(j)})]$$

Second term — $z^{(j)} = W_{hh} h^{(j-1)} + \dots$ is linear in $h^{(j-1)}$, so:

$$\frac{\partial z^{(j)}}{\partial h^{(j-1)}} = W_{hh}$$

Its transpose appears because we're computing the Jacobian (mapping gradient vectors), giving W_{hh}^T .

Combining:

$$\frac{\partial h^{(j)}}{\partial h^{(j-1)}} = W_{hh}^T \cdot \text{diag}[f'(W_{hh} h^{(j-1)} + W_{hx} x^{(j)})]$$

Why the first term is a diagonal matrix

The key is that f is applied **elementwise**. That means each output component $h_i^{(j)}$ depends **only** on $z_i^{(j)}$, not on any other $z_k^{(j)}$.

So when you take the partial derivative:

$$\left[\frac{\partial h^{(j)}}{\partial z^{(j)}} \right]_{ik} = \frac{\partial h_i^{(j)}}{\partial z_k^{(j)}}$$

- When $i = k$: this equals $f'(z_i^{(j)})$ ✓
- When $i \neq k$: this equals 0, because h_i doesn't depend on z_k ✓

So the full Jacobian matrix looks like:

$$\frac{\partial h^{(j)}}{\partial z^{(j)}} = \begin{bmatrix} f'(z_1) & 0 & \cdots & 0 \\ 0 & f'(z_2) & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & f'(z_n) \end{bmatrix}$$

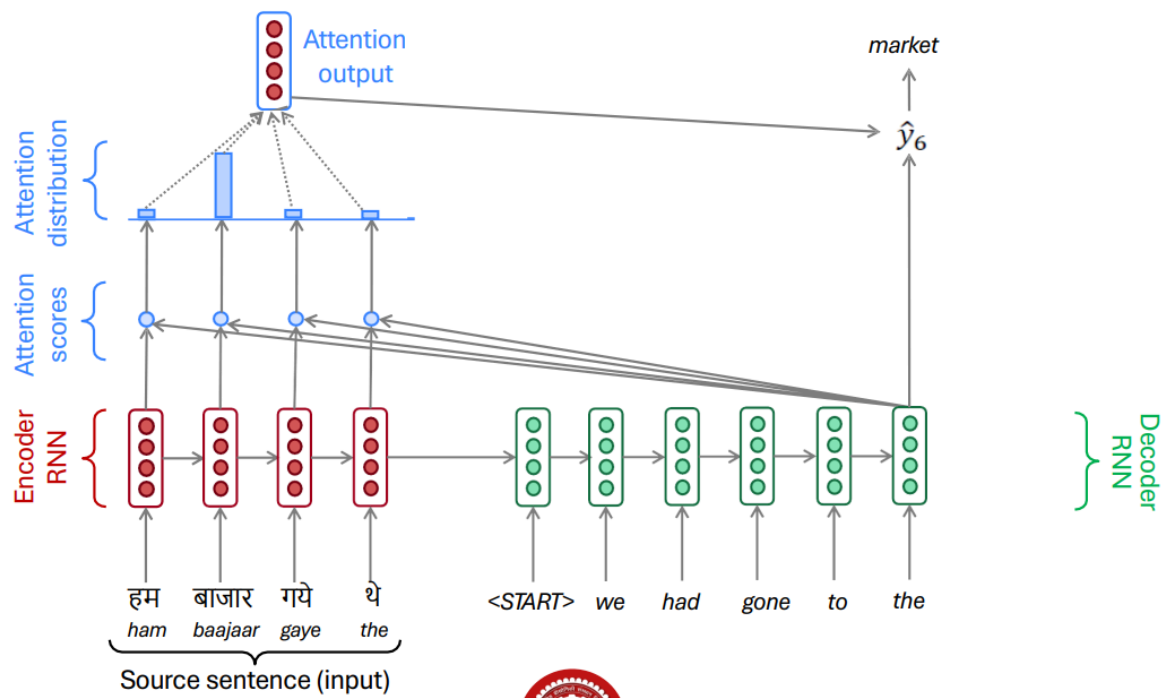
Which is exactly what $\text{diag}[f'(z^{(j)})]$ means — a matrix with the derivatives on the diagonal and zeros everywhere else.

5. Vanishing / Exploding Gradient

Your matrix			Raised to the 5th power			Raised to the 10th power			Raised to the 15th power			Raised to the 20th power		
	A ₁	A ₂		A ₁	A ₂		A ₁	A ₂		A ₁	A ₂		A ₁	A ₂
1	0.1	0.5	1	0.13056	0.1408	1	0.0448004096	0.044695552	1	0.014659528687616	0.01466060242944	1	0.004803846016346358	0.00480383502123008
2	0.7	0.3	2	0.19712	0.18688	2	0.0625737728	0.0626786304	2	0.020524843401216	0.020523769659392	2	0.006725369029722112	0.00672538002483839

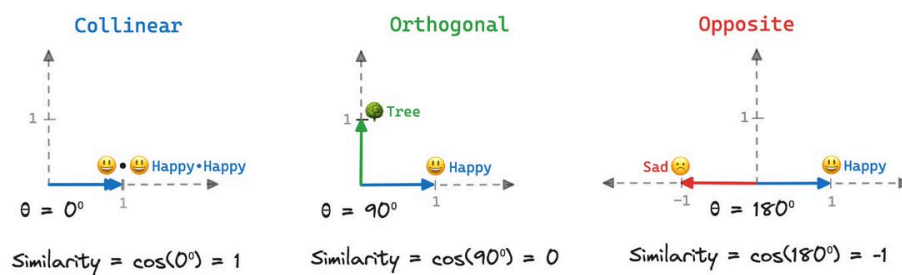
Your matrix			Raised to the 5th power			Raised to the 10th power			Raised to the 15th power			Raised to the 20th power		
	A_1	A_2		A_1	A_2		A_1	A_2		A_1	A_2		A_1	A_2
1	1.1	1.5	1	80.66816	80.6784	1	13884.2234945536	13884.223389696	1	2389525.9477674253	2389525.947768499	1	411246209.0663599	411246209.0663599
2	1.7	1.3	2	91.43552	91.42528	2	15735.4531749888	15735.4532798464	2	2708129.4074709658	2708129.407469892	2	466079036.9418745	466079036.9418745

6. Seq-to-Seq with Attention



7. Cosine Similarity Intuition

A Geometric intuition to vector similarity! 🚀



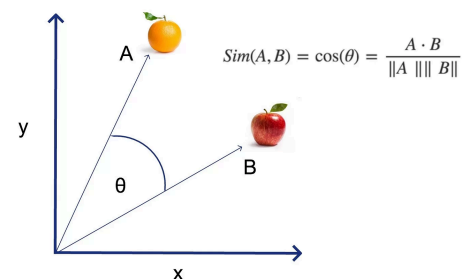
Collinear: The vectors/words are same!

Orthogonal: The vectors/words are totally un-related!

Opposite: The vectors/words are exact opposites!

@akshay_pachar

Cosine Similarity



8.